

Source System



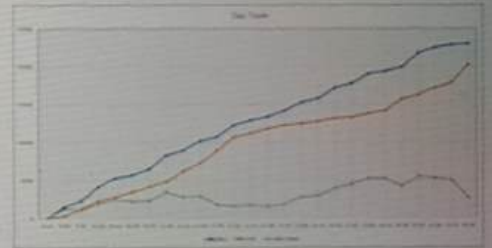
A Large Stock Brokerage Company



Source System



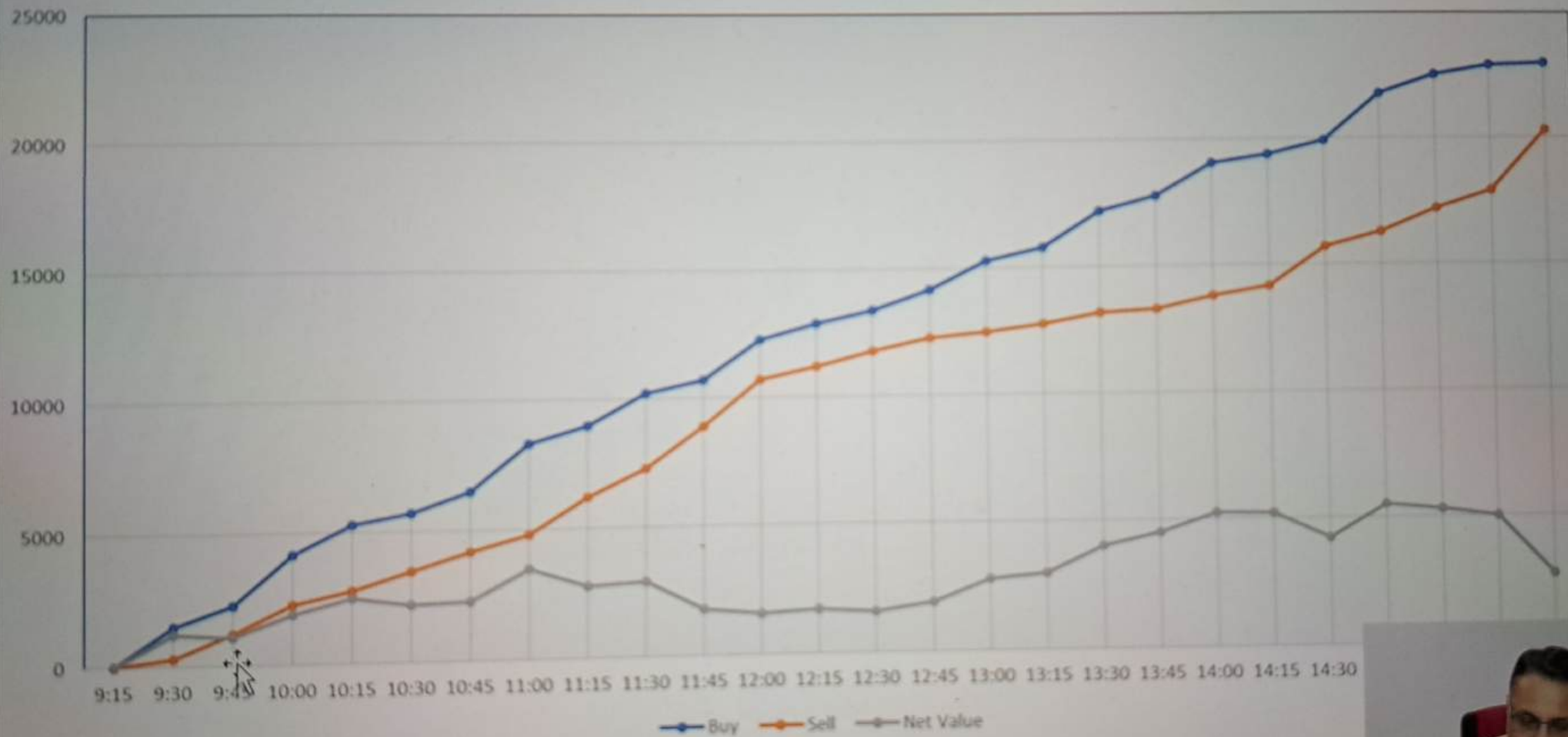
Consumer Application



Trade Summary Chart



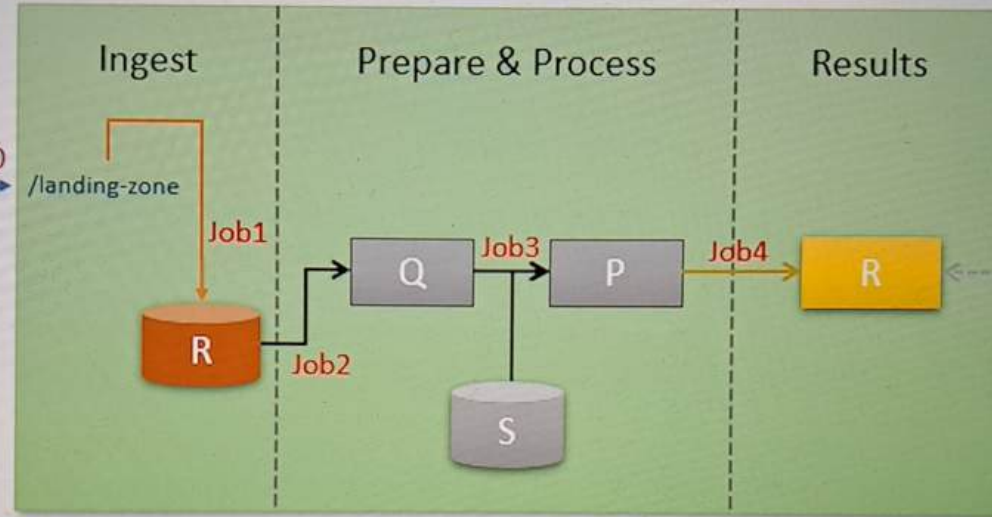
Day Trade



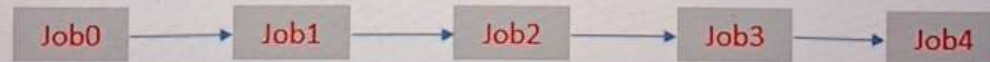
Source System



Data Engineering Platform



Data Pipeline (Workflow)



What is the pipeline frequency?

Consumer Application

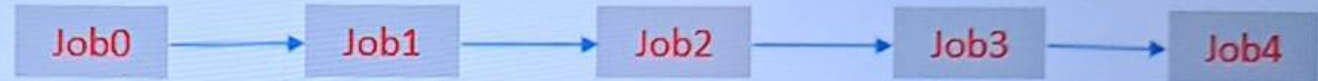


Trade Summary Chart



Data Pipeline (Workflow)

Job frequency?



1. Monthly
2. Weekly
3. Daily
4. Every Hour

Enough time between iterations simplifies the problem.

Back Pressure

1. Every 15 minutes
2. Every minute

1. Startup time?
2. Processing time and volume?
3. Cleanup/Housekeeping time?

3. Every second
4. As soon as possible (milliseconds)

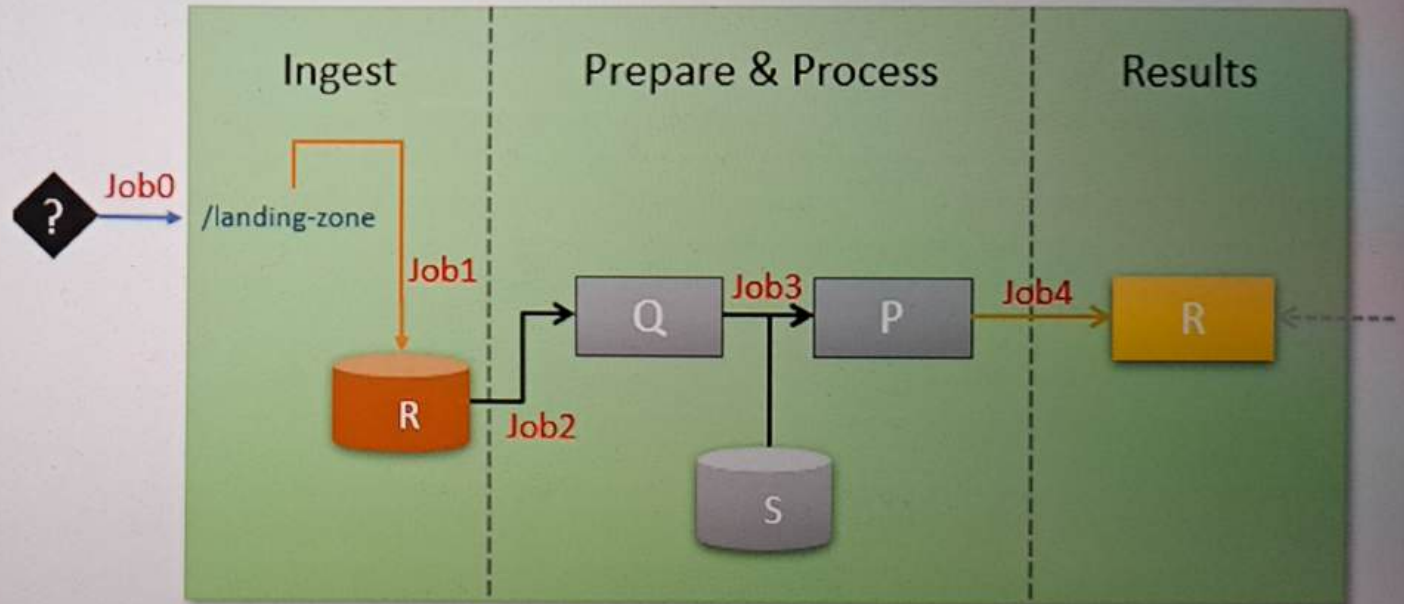
1. How to schedule for co listening



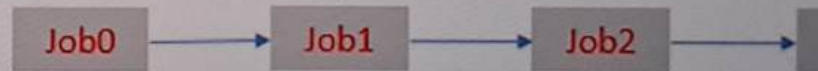
Improve job performance and avoid backpressure – Process only incremental data.

How do we process incremental data?

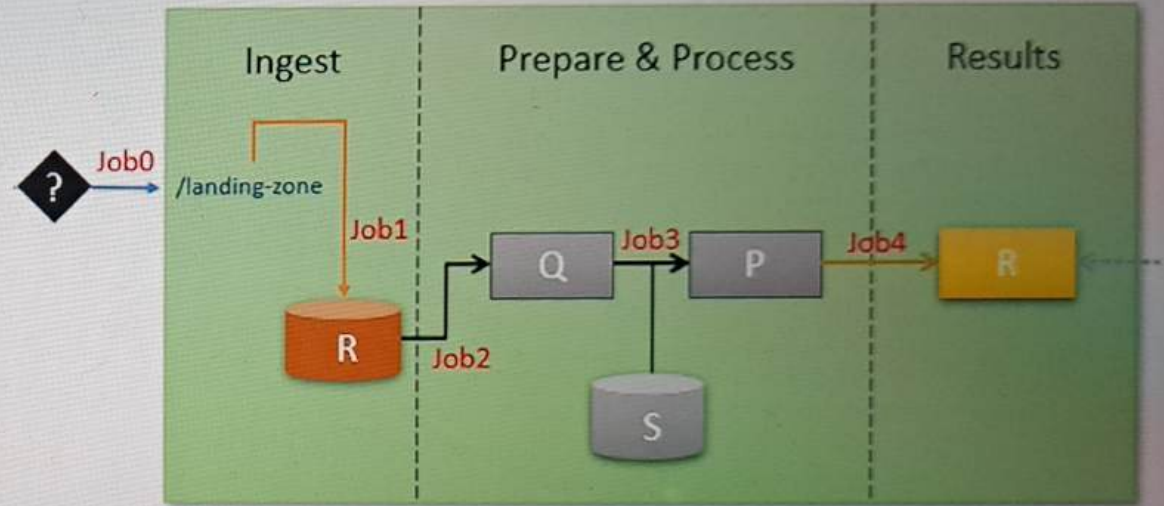
What is new? – Checkpoint



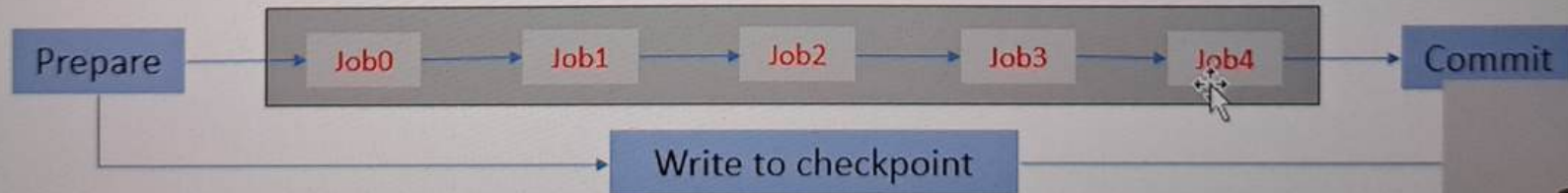
Data Pipeline (Workflow)



How to handle failure?



How to implement checkpoints?



How do you handle late records? – Updating state

Iteration 1 @ 10.00

New records till 10.00 AM

Time	Sale	Purchase
9.32	1200	0
9.33	900	0
9.35	0	800
9.45	1300	0
9.52	0	2200
	3400	3000

Aggregate calculated at 10.00 AM

Time	Total Sales	Total Purchase
10.00	3400	3000

 ✓

Iteration 2 @ 10.00

New records till 10.30 AM

Time	Sale	Purchase
9.59	400	0
10.03	1300	0
10.05	0	800
10.02	0	600
10.15	900	0
10.18	0	1500
10.26	500	0
	2700	2900

Aggregate calculated at 10.30 AM

Time	Total Sales	Total Purchase
10.00	3400	3000
10.30	2700	2900

 ✗ ✓

APACHE
SparkTM
Structured Streaming

Backpressure – How to handle backpressure

Workflow – How to schedule the job for continuous listening

Checkpoints – Creating and managing checkpoints for incremental processing

Fault tolerance – Transactional Checkpointing

State Management – Managing aggregate states and updating states for late-arriving data

